

Predicting Corporate Bond YTM Changes Using Machine Learning

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1. Objective

The objective of this project is to predict corporate bond yield to maturity (YTM) changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators.

By applying various machine learning regression models such as ElasticNet, Boosting and Multi-layer Perceptron, we aim to improve prediction accuracy compared to traditional linear regression models.

In addition, we used the change direction of YTM movements as a classification target since there are many outliers in the YTM change values that may affect regression performance. We applied classification models such as Logistic Regression, ElasticNet Classifier, and Boosting Classifier to predict the direction of YTM changes.

2. Methodology

2.1 Data Description

We will utilize data from Wharton Research Data Services (WRDS) to construct our dataset. The data frequency is monthly and data range is from July 2002 to February 2025.

We totally have 50 features from four categories: bond-level features, firm-level features, macroeconomic features, and industry features. The details of the target variable and features are described below.

2.1.1 Target Variable The target variable is the corporate bond yield to maturity (YTM) change, defined as the difference in YTM over a one-month horizon:

$$\Delta YTM_i = YTM_{i,t} - YTM_{i,t-1}$$

where $YTM_{i,t}$ is the yield to maturity of bond i at the end of month t .

For classification, the target variable is defined as:

$$YTM_Direction_i = \begin{cases} up, & \text{if } \Delta YTM_i > 1e-6 \\ neutral, & \text{if } \Delta YTM_i \approx 0 \\ down, & \text{if } \Delta YTM_i < -1e-6 \end{cases}$$

The Data is from the “WRDS Bond Returns” database, which provides monthly bond yields and other bond characteristics, sourced from TRACE and Mergent FISD. The dataset has been cleaned and more suitable for analysis.

We only consider senior bonds and exclude defaulted bonds. Additionally, the yields exceeding $[-1, 1]$ are treated as outliers and removed from the dataset.

Total number of observations meeting the criteria is approximately 1760852, covering 74191 unique bonds and 3187 unique firms over the sample period.

After merging with firm-level and macroeconomic data, the final dataset contains 12117 unique bonds.

The data descriptive statistics of the yield to maturity (YTM) changes are as follows:

```
count      854769.000000
mean        -0.000005
std         0.014440
min         -0.980800
25%         -0.001840
50%         -0.000030
75%         0.001710
max         0.979029
Name: ytm_chg, dtype: float64
```

The number of observations for each YTM change direction category is as follows:

```
down      435398
up        399205
neutral    20166
Name: count, dtype: int64
```

2.1.2 Features

1. Key Columns

- Bond Identifier: CUSIP
- Firm Identifier: PERMNO
- Date: End of Month

We have checked the missingness and uniqueness of these key columns.

2. Bond-level features

- Time to Maturity (tmt): the remaining time to maturity of the bond in years.
- Coupon Rate (coupon): the annual coupon rate of the bond.
- t_spread: the weighted average bid-ask spread of the bond which measures liquidity.
- return_eom: bond return of last month.

- Credit Ratings: One-hot encoded variables for ratings from AAA to D (rating_A, rating_AA, ..., rating_D). The ratings are weighted averages ratings from WRDS Bond Returns database rather than S&P, Moody's, or Fitch alone.
- Rating Change Indicators: Binary variables indicating whether there was an upgrade or downgrade in the bond's credit rating in the past month (upgrade, downgrade).

Here, we used lagged values of features except time to maturity and coupon rate to avoid data leakage.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
tmt	3.56e+6	6.35	8.67	0.00	1.61	3.54	6.89	102.13
coupon	3.56e+6	0.03	0.03	0.00	0.00	0.03	0.06	0.34
t_spread	1.79e+6	0.01	0.01	0.00	0.00	0.00	0.01	2.00
yield	3.39e+6	0.01	0.13	-1.00	0.00	0.03	0.05	1.00
ret_eom	3.42e+6	0.01	1.18	-1.00	0.00	0.00	0.01	2090.0
rating_A	3.42e+6	0.20	0.40	0.00	0.00	0.00	0.00	1.00
rating_AA	3.42e+6	0.04	0.21	0.00	0.00	0.00	0.00	1.00
rating_AAA	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_B	3.42e+6	0.03	0.16	0.00	0.00	0.00	0.00	1.00
rating_BB	3.42e+6	0.04	0.20	0.00	0.00	0.00	0.00	1.00
rating_BBB	3.42e+6	0.23	0.42	0.00	0.00	0.00	0.00	1.00
rating_C	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
rating_CC	3.42e+6	0.00	0.03	0.00	0.00	0.00	0.00	1.00
rating_CCC	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_D	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
upgrade	3.42e+6	0.00	0.04	0.00	0.00	0.00	0.00	1.00
downgrade	3.42e+6	0.00	0.05	0.00	0.00	0.00	0.00	1.00

3. Firm-level features (yutung)

We incorporate firm-level information from both daily stock data (CRSP) and quarterly accounting data (Compustat). These datasets provide market-based signals and fundamental characteristics of the issuing firms. We then construct additional financial ratios and growth indicators used as predictors.

(1) Stock Market Features (Monthly Aggregated)

- ret — Monthly stock return (accumulated from daily returns).
- vol — Monthly return volatility (std of daily log returns).
- dvol — Dollar trading volume (price $\times$ shares traded).
- turnover — Trading turnover (shares traded $\div$ shares outstanding).
- bidask — Bid-ask spread based on daily midquotes.
- numtrades — Number of trades in the month.
- mktcap — Month-end market capitalization.
- price_mean — Average daily stock price.

These features reflect firm-level market performance, trading activity, and liquidity conditions.

(2) Accounting-Based Derived Indicators (Monthly Filled)

- `log_atq` — Log total assets (firm size proxy).
- `lev_total` — Leverage ratio = total liabilities $\div$ total assets.
- `equity_ratio` — Equity-to-assets ratio.
- `roa` — Profitability = net income $\div$ total assets.
- `profit_margin` — Net profit margin = net income $\div$ revenue.
- `int_coverage` — Ability to service debt = operating income $\div$ interest expense.
- `mkt_cap` — Accounting-based market cap = `prccq` $\times$ `cshoq`.
- `market_to_book (m2b)` — Valuation ratio = (market cap $\div$ common equity). These variables summarize capital structure, profitability, solvency, and valuation.

(3) Growth Indicators (QoQ)

- `atq_growth` — Asset growth (decomposition of balance-sheet expansion).
- `revtq_growth` — Revenue growth (sales momentum).
- `niq_growth` — Net income growth (profitability momentum). These features capture firm momentum and fundamental acceleration, which are known predictive signals in credit and bond return literature.

Variable	Mean	Std	Min	25%	50%	75%	Max
<code>stock_ret</code>	0.00081	0.1162	-0.3446	-0.052	0.00206	0.05164	0.3881
<code>stock_vol</code>	0.02420	0.01729	0	0.01106	0.01978	0.03373	0.07639
<code>dvol</code>	4.720e8	1.338e9	0	3.564e6	2.814e7	2.269e8	9.083e9
<code>price_mean</code>	27.89	33.39	0.4247	7.503	17.41	34.67	202.7
<code>turnover</code>	10.55	20.04	0.1046	2.171	5.032	10.39	151.0
<code>bidask</code>	0.03700	0.03154	0.00136	0.01481	0.02762	0.04922	0.1571
<code>numtrades</code>	2.353e4	7.754e4	0	0	0	5060	5.375e5
<code>stock_mktcap</code>	2.558e6	7.306e6	2612	6.575e4	2.860e5	1.336e6	5.040e7
<code>log_atq</code>	5.966	2.271	1.059	4.265	5.935	7.575	11.88
<code>lev_total</code>	0.5453	0.2667	0.04291	0.3332	0.5433	0.7454	1.437
<code>equity_ratio</code>	0.4682	0.2518	0.03758	0.2655	0.4584	0.6722	0.9650
<code>roa</code>	-0.011	0.06202	-0.4176	-0.001	0.00419	0.01634	0.1046
<code>profit_margin</code>	-0.067	0.5065	-4.911	-0.032	0.03624	0.09573	0.8525
<code>int_coverage</code>	7.883	10.21	-9.713	2.324	4.903	9.881	78.47
<code>mkt_cap</code>	3631	2.366e4	0.0150	55.60	249.9	1257	4.330e6
<code>m2b</code>	2.865	3.123	0.2618	1.144	1.846	3.275	26.64
<code>atq_growth</code>	0.0306	0.2174	-1.000	-0.023	0.00585	0.04315	4.966
<code>revtq_growth</code>	0.0594	0.3745	-1.000	-0.058	0.01596	0.1119	5.000
<code>niq_growth</code>	-0.111	1.158	-5.000	-0.538	-0.033	0.2378	5.000

4. Macroeconomic features (`yuxi`)

- Equity Market Return (sp500):
Monthly level of a broad U.S. equity index (S&P 500), used to capture overall stock market conditions and risk appetite.
- Short-Term Interest Rate (ir3m):
Monthly average of the 3-month Treasury yield, serving as the short end of the risk-free term structure and a proxy for monetary policy stance.
- Long-Term Interest Rate (ir10y):
Monthly average of the 10-year Treasury yield, capturing long-term discount rates relevant for bond valuation.
- Implied Volatility (vix):
Monthly average of the VIX index, measuring market-implied equity volatility and shifts in aggregate risk aversion.
- Real Economic Activity (gdp):
Real GDP level (or growth, depending on transformation) aligned to the month of observation, summarizing aggregate economic conditions.
- Inflation (cpi):
Consumer Price Index level (or inflation rate) matched to each month, capturing the inflation environment relevant for nominal yields and real required returns.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
SP500	335	2185.71	1418.34	735.09	1191.41	1454.60	2772.33	6840.20
IR3M	335	2.07	2.00	0.00	0.12	1.51	4.21	6.15
IR10Y	335	3.44	1.37	0.54	2.30	3.53	4.49	6.67
VIX	335	20.29	7.82	9.51	14.45	18.43	24.44	59.89
GDP	328	17764.90	2911.28	12703.74	15670.88	17035.11	20070.68	23770.98
CPI	333	227.69	42.89	162.00	191.70	227.17	252.56	324.37

5. Industry features (yuxi)

- Sector ETF Price and Return (price, return):
For each bond, we link a sector ETF based on its GICS sector, and use the ETF's monthly price (**price**) and monthly return (**return**) to proxy for sector-level performance and sector-specific shocks that may not be fully captured by firm-level variables.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Price	3128	44.03	39.40	4.57	18.23	29.43	54.97	300.68
Return	3117	0.008	0.055	-	-	0.0110	0.0390	0.3076
				0.3437	0.0217			

2.2 Feature Construction (Yutung)

After preparing the bond-level, firm-level, sector-level, and macroeconomic datasets, we align all information at a monthly frequency and construct the final panel used for modeling. This stage focuses on dataset merging, temporal alignment, lagging to avoid look-ahead bias, missing-value handling, and feature normalization.

1. Merging Procedure

We combine the datasets in the following order to maintain consistent identifiers and time alignment:

- Stock + Fundamentals (PERMNO, date): CRSP monthly stock aggregates are merged with Compustat monthly-filled accounting fundamentals using the common firm identifier (PERMNO) and month-end date.
- Add Macroeconomic Variables (date): Monthly macro data are merged using the calendar month-end date.
- Add Sector ETF Information (industry code): Each firm is mapped to its sector ETF using global industry classification, and monthly ETF returns/prices are merged accordingly.
- Merge with Bond Data (CUSIP root, date): Bonds are linked to issuers via the first six digits of CUSIP (issuer6). Only issuer-month pairs that appear in both datasets are retained to ensure valid alignment.

2. Missing-Value Handling

After merging, missing values are handled according to variable type:

Categorical Variables:

- Credit rating dummies (AAA...D)
- Rating transition indicators
- Compustat data-status flag (costat) Here we use cross-sectionally monthly median to impute missing values, preserving the categorical distribution across firms each month.

Numeric Variables: We also use cross-sectionally monthly median to impute missing values for numeric variables, then for remaining NaNs, we set them to zero.

3. Normalization

(1) Rank-Normalized Features

- Includes stock features, fundamentals, derived ratios, growth variables, and bond-level numeric predictors.

- Each month, variables are transformed using:

$$\text{scaled_rank} = 2 \times \frac{\text{rank}}{N} - 1$$

producing values in $[-1, 1]$.

- For binary variables (e.g., upgrade/downgrade), ranks are assigned such that 1 maps to 1 and 0 maps to -1 .
- (2) Not Normalized (kept in raw form) includes macroeconomic indicators and yield-curve variables:
- sp500_ret, gdp_gr, cpi_infl
 - ir3m_chg, ir10y_chg
 - vix_chg
 - s3m, term_spread

These variables are identical across all firms in a given month; rank-normalizing them collapses them into constants, eliminating their informational content. Therefore, they are kept in raw form to preserve meaningful macro signals.

2.3 Data Splitting

In this project, we used a rolling window forecasting approach similar to our homework. At each year, we used a time-series split to create training, validation, and test sets. The training set consisted of 10 years of data, the validation set and test set each consisted of 1 year of data. When tuning is not involved, the validation set was combined with the training set to train the final model. For example, to predict YTM changes in 2012, we used data from 2002 to 2011 as the training set, data from 2012 as the validation set, and data from 2013 as the test set. Then, to predict YTM changes in 2013, we used data from 2003 to 2012 as the training set, data from 2013 as the validation set, and data from 2014 as the test set. This process was repeated until we predicted YTM changes in 2024.

2.4 Modeling Framework

2.4.1 Regression Models

We implemented and compared the following machine learning models for regression task:

Model	Description
Linear Regression	Baseline linear model
ElasticNet	Combination of L1 and L2
Boosting (LightGBM)	Ensemble of weak learners
Multi-layer Perceptron	Feedforward neural network

Model	Description
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Here, for the ElasticNet model, we tuned the hyperparameters `alpha` and `l1_ratio` using 10-fold cross-validation on the training set. However, we only tuned once based on the first training set and used the same hyperparameters for all other training sets to reduce computation time. This approach was applied to all tuning processes in this project. For the Boosting model, we tuned the hyperparameters using a grid search.

We also implemented a Stacked Regressor that combines predictions from Linear Regression, ElasticNet, and LightGBM models using linear regression as the meta-model to improve overall performance.

For the Multilayer Perceptron (MLP), we use the same rank-normalized predictors as in the other regression models (bond-level, firm-level, and sector-level features), together with the raw macro variables. The MLP is implemented as a small fully connected feed-forward network with ReLU activation functions and a single linear output node, trained with mean squared error loss and the Adam optimizer. For each rolling window, we re-estimate the MLP on the corresponding training period and evaluate it on the test year, using an internal hold-out split within the training window for early stopping. Thus, the MLP is directly comparable to the other regression models in terms of the rolling-window evaluation design, although it is considerably more computationally intensive and we keep the network architecture relatively small to control training time.

The MLP is not part of the stacked regressor due to its separate training procedure and high computational cost.

Notice that in each load, we normalized the macroeconomic variables based on the training set statistics. That’s because these variables are identical across all firms in a given month, normalizing them using the entire dataset would lead to data leakage and normalizing them monthly would collapse them into constants, eliminating their informational content.

2.4.2 Classification Models

For classification task, we implemented and compared the following models:

Model	Description
Logistic Regression	Linear classification model
ElasticNet Classifier	Combination of L1 and L2
Boosting Classifier (LightGBM)	Ensemble of weak learners

Here, we also tuned hyperparameters for ElasticNet Classifier and Boosting Classifier using similar approaches as in regression task.

For the Stacked Classifier, we combined predictions from Logistic Regression, ElasticNet Classifier, and LightGBM Classifier using logistic regression as the meta-model.

2.5 Model Evaluation

We used a rolling window evaluation approach and calculated performance metrics on the test set for each year from 2012 to 2024. Then, the metrics were averaged to assess overall model performance.

For regression models, we evaluated performance using:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Median Absolute Error (MedAE)
- R^2

For classification models, we evaluated performance using:

- Accuracy: Overall correctness of the model.
- Precision: Proportion of positive identifications that were actually correct.
- Recall: Proportion of actual positives that were identified correctly.

3. Results and Discussion

3.1 Regression Results

The table below summarizes the performance of different regression models on the test set:

Model	MSE	MAE	MedAE	R^2
Elastic Net Regression	0.000093	0.003167	0.002032	0.005956
LGBM Regressor	0.000084	0.002817	0.001607	0.085135
Linear Regression	0.000093	0.003171	0.002036	0.004982
Stacked Regressor	0.000084	0.002841	0.001607	0.063947
Multilayer Perceptron	0.000262	0.005025	0.002683	0.013985

The Elastic Net Regression model did not significantly outperform the Linear Regression model, indicating that regularization may not provide substantial benefits in this context. The LGBM Regressor achieved the lowest MSE and highest R^2 , suggesting that ensemble methods can better capture complex relationships in the data. The Stacked Regressor also performed well, leveraging the strengths of multiple models but did not surpass the LGBM Regressor.

The multilayer perceptron (MLP) underperforms all other models in the rolling experiment, with noticeably higher MSE/MAE and only a modest $R^2 \approx 0.014$. This suggests that, given our current feature set and simple network architecture, a feed-forward neural network does not extract additional predictive structure

beyond what linear models and tree-based ensembles already capture, and may even be more prone to fitting noise.

3.2 Classification Results

As we mentioned before, we defined the target variable for classification as the direction of YTM changes (up, neutral, down) to mitigate the impact of outliers. The table below summarizes the performance of different classification models on the test set:

Model	Accuracy	Precision	Recall
Elastic Net Classification	0.682760	0.694127	0.682760
LGBM Classifier	0.713273	0.722421	0.713273
Logistic Regression	0.682736	0.694145	0.682736
Stacked Classifier	0.706753	0.718828	0.706753

In classification tasks, both the Elastic Net Classifier and Logistic Regression models exhibited similar performance, which is consistent with the regression results. The LGBM Classifier outperformed the other models, achieving the highest accuracy, precision, and recall. The Stacked Classifier also showed strong performance, although it did not exceed the LGBM Classifier.

4. Conclusion

In this project, we explored various machine learning models to predict corporate bond yield to maturity changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators. Our findings indicate that ensemble methods like LightGBM outperform traditional linear models in both regression and classification tasks. Regularization techniques such as ElasticNet did not yield significant improvements over linear models in this context. However, the improvement was modest, suggesting that further enhancements such as more accurate data, additional features, or alternative modeling approaches may be necessary to achieve substantial gains in predictive accuracy. textual disclosures, market microstructure signals), or alternative modeling frameworks that more directly account for regime shifts and rare credit events.

Appendix:

A.1 Implementation Details

1. Software and Libraries
 - CPU: MacBook Pro (Apple M4 Pro)
 - Programming Language: Python 3.13
 - Libraries: pandas, numpy, scikit-learn, lightgbm, tensorflow/keras
 - Code Repository: Github [https://github.com/QinqinAndMacaulayCat/ml_project/tree/main]

2. Computation Time For Each Model (Approximate)

Model	Tuning	Time for Training and Prediction
Linear Regression		3.9 seconds
Elastic Net Regression		1 minute 10 seconds
LGBM Regressor	31 minutes 41.4 seconds	58.3 seconds
Stacked Regressor		6 minutes 50 seconds
Logistic Regression		1 minute 30 seconds
Elastic Net	83 minutes 59.8 seconds	23 minutes 40.7 seconds
Classification		
Boosting Classifier	114 minutes 48.8 seconds	1 minute 27.9 seconds
Stacked Classifier		107 minutes 11.9 seconds
Multilayer Perceptron		3 minutes 16 seconds

B.1 Classification Confusion Matrix

The confusion matrixes are as follows:

1. Logistic Classification

actual \ predicted	up	down	neutral
up	155404	103686	264
down	54596	205732	322
neutral	3986	6213	1259

2. Elastic Net Classification

actual \ predicted	up	down	neutral
up	155442	103662	250
down	54610	205719	321
neutral	3990	6218	1250

3. LGBM Classifier

actual \ predicted	up	down	neutral
up	174264	84514	576
down	58838	201267	545
neutral	2317	3105	6036

4. Stacked Classifier

actual \ predicted	up	down	neutral
up	176566	82735	53
down	62485	198119	46
neutral	3147	4741	3570